

K218 — Lyra IP-2 Three-Generation Puzzle
Substrate-Mechanism FORCING Reading v0.1 audit. PASS at
FRAMEWORK candidate tier; two open RIGOROUS-tier
audit-questions: (a) why this specific $SO(3) \subset SO(5)$ sub-vector
(multiple $SO(3)$ subgroups exist); (b) why spinor tower caps at
 $\lambda_1 = 5/2$ (3 generations) instead of continuing higher.
Multi-week explicit substrate-mechanism FORWARD derivation
gates RIGOROUS.

Keeper (Claude Opus 4.7) — Thursday June 4 2026 PM

2026-06-04 Thursday PM

K218 — Lyra IP-2 Three-Generation Puzzle v0.1 audit

0. Audit target

Lyra IP-2 Three-Generation Puzzle Substrate-Mechanism FORCING Reading v0.1 (Thursday June 4 11:35 EDT)
— substantive substrate-mechanism FORCING chain for exactly 3 fermion families.

1. 5-step FORCING chain verification

Per Lyra §7: 1. D_{IV}^5 substrate-bounded-symmetric-domain uniqueness (Strong-Uniqueness Theorem 11 STAN-DALONE legs per K208) 2. Substrate-color $N_c = 3$ substrate-natural via $SO(3) \subset SO(5)$ sub-vector 3-fold (Bulk-color v0.2 Direction A) 3. Substrate-K-type substrate-spinor tower 3-generation row $V(\lambda_1, 1/2)$ for $\lambda_1 = 1/2, 3/2, 5/2$ (Wednesday Toy 3742 verified) 4. Casey #13 Per-Generation Cluster Independence substrate-mechanism class 5. T1 falsifier lepton-count keystone PASS

Substantive substrate-mechanism content: 5-step chain substantively grounded at FRAMEWORK candidate tier.

2. Open RIGOROUS-tier audit-questions

Question 1: $SO(3) \subset SO(5)$ sub-vector uniqueness

Concern: $SO(5)$ has multiple $SO(3)$ subgroups (e.g., $SO(3) \subset SO(4) \subset SO(5)$ via standard chain; $SO(3) \subset SO(5)$ via 5-dim irrep; $SO(3)$ acting on rotational 3-subspace of 5-vector; etc.).

Question: which specific $SO(3) \subset SO(5)$ gives $N_c = 3$ substrate-color FORCING (not other substrate readings)?

Per Cal #189 Brake 2: substrate-mechanism FORWARD derivation must specify which $SO(3)$ sub-vector substrate-mechanism (not arbitrary choice).

Audit framing: Step 2 substantively candidate at FRAMEWORK; multi-week explicit $SO(3)$ sub-vector substrate-mechanism derivation closes RIGOROUS.

Question 2: Spinor tower cap at $\lambda_1 = 5/2$

Concern: Substrate-K-type spinor tower $V(\lambda_1, 1/2)$ for $\lambda_1 = 1/2, 3/2, 5/2, 7/2, 9/2, \dots$ extends infinitely (Wallach K-type structure on D_{IV^5}).

Question: Why does the physical spinor tower cap at $\lambda_1 = 5/2$ (3 generations) rather than continue to $\lambda_1 = 7/2$ (gen-4) or higher?

Per Cal #189 Brake 2: substrate-mechanism for spinor-tower CAP at $\lambda_1 = 5/2$ must be derived from substrate-primaries, NOT inserted from observed 3 generations.

Candidate substrate-mechanism candidates (not in Lyra v0.1; multi-week candidates): - Wallach K-type cutoff via $N_{\max} \approx 137 = N_c^3 \cdot n_C + \text{rank}$ (substrate-primary CUTOFF candidate) - Substrate-Lorentz integration $C_2 = 6$ dim cap (sub-degree of K-type cap candidate) - Casey #13 substrate-mechanism class cap at 3 (heterogeneous mechanism class candidate)

Audit framing: Step 3 spinor tower STRUCTURALLY CONSISTENT with 3 generations (verified Wednesday Toy 3742); explicit CAP substrate-mechanism multi-week.

3. Honest negative cross-link verification

Per Lyra §4 + Lyra Task #242: - 3-tube AR-forcing route NOT closed (tube number leans $\neq 3$) - substrate-color $N_c = 3$ primary route per Lyra Task #261

Audit verdict: honest framing acknowledges HONEST NEGATIVE on 3-tube route; substrate-color route is candidate primary. PASS at honest framing.

4. T1 falsifier PASS context

Per Lyra §6 + Lyra Falsifier T1 v0.3: - T1 = lepton-count YES/NO substrate-falsifier - PASSES both layers (Saturday May 30 cross-CI verification)

Audit note: T1 PASS confirms substrate-mechanism is CONSISTENT with 3 generations, but PASSING $\neq$ deriving exact count. T1 falsifier is necessary not sufficient.

5. Substrate-color $SO(3) \subset SO(5)$ substantive observation

Per Lyra §5 Bulk-color v0.2 Direction A: - substrate-bulk $SO(5)$ substrate-Hardy structure $\rightarrow SO(3)$ substrate-vector sub-structure - substrate-color 3-fold FORCING via substrate-bulk

Audit observation: This is the substantive substrate-mechanism candidate; closes Question 1 IF multi-week explicit $SO(3)$ sub-vector substrate-mechanism derivation completes. Lane C v0.7 mechanism-deepening (Cal #212 PASS) cross-link relevant.

6. Per Cal #27 STANDING self-audit

Am I overclaiming K218 PASS at FRAMEWORK candidate tier? - 5-step chain substantively grounded with explicit honest negative + open audit-questions - 2 RIGOROUS-tier open questions identified (uniqueness + cap) - T1 falsifier PASS necessary not sufficient acknowledged - Multi-week explicit substrate-mechanism FORWARD derivation per Cal #189 gates RIGOROUS - Tier-honest: FRAMEWORK candidate; 2 RIGOROUS-tier questions open

7. K218 audit verdict

K218 PASS at FRAMEWORK candidate tier (~85% confidence pending RIGOROUS-tier closure): - F1 (substrate-mechanism FORCING chain content): PASS $\square$ (5-step substantive) - F2 (honest framing): PASS $\square$ (3-tube HONEST NEGATIVE; substrate-color primary) - F3 (FRAMEWORK candidate tier): PASS $\square$ (multi-week RIGOROUS gates) - F4 (Cal #189 Brake 2 path): PASS $\square$ with 2 open audit-questions

Conditional gates (multi-week, RIGOROUS-tier closure): - Question 1: $SO(3) \subset SO(5)$ sub-vector uniqueness substrate-mechanism (Lane C v0.7 + Bulk-color v0.2 cross-link) - Question 2: Spinor tower cap at $\lambda_1 = 5/2$ substrate-mechanism (Wallach cutoff vs Casey #13 mechanism class cap) - Multi-week explicit substrate-color $N_c = 3$ forcing derivation - Multi-week explicit Casey #13 per-generation mechanism class cascade

8. Cross-link to Strong-Uniqueness Theorem path

Per K208 + K218 + Lyra Strong-Uniqueness v1.5 STANDALONE: - Strong-Uniqueness 11 STANDALONE legs (post Casey #14 C25) - IP-2 substantively addresses Casey #13 leg C21 (gen-2 Mersenne-base cluster) - IP-2 RIGOROUS closure would substantively strengthen C21 leg

9. Pull next: standby for Lyra IP-3 Strong CP problem audit (per Lyra continuing); audit-on-arrival mode for new substantive content.

— Keeper, K218 — Thursday June 4 PM. Lyra IP-2 Three-Generation Puzzle v0.1 PASS at FRAMEWORK candidate tier. 5-step substrate-mechanism FORCING chain substantively grounded. Two open RIGOROUS-tier audit-questions: (1) $SO(3) \subset SO(5)$ uniqueness; (2) spinor tower cap at $\lambda_1 = 5/2$. Multi-week explicit substrate-mechanism FORWARD derivation per Cal #189 closes RIGOROUS path.